

Abdelrahman Ahmed

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Education

American University of Sharjah

Sharjah, UAE

Bachelor of Science in Computer Science | Dean's List, Spring 2026

Expected May 2027

Relevant Coursework: Data Structures, Computer Networks, Web Applications & Development, Database Systems, Operating Systems, Cybersecurity

Independent Study: MIT 6.S184 Flow Matching & Diffusion Models; transformers and attention; graph neural networks; optimal transport

Technical Skills

Languages: Python, Java, JavaScript, C++, SQL, HTML, CSS

Machine Learning: PyTorch, NumPy, transformers, graph neural networks, optimal transport

Developer Tools: Git, Linux, Jupyter, Jan AI

Databases: MySQL, SQLite

Systems & Networking: TCP/IP, Subnetting, Routing and Switching, Raspberry Pi, Embedded Linux

Experience

Research Assistant – Machine Learning & Computer Vision

Jun 2026 – Aug 2026

The University of British Columbia

Kelowna, BC, Canada

- Research transformer-based visual object tracking under Dr. Mohamed Shehata, extending a one-stream tracker with a graph neural network branch that improves information flow between template and search tokens
- Proposed and implemented a new normalization technique for the graph branch in PyTorch, built as a configurable module so existing baselines remain reproducible and comparisons stay controlled
- Implemented graph attention, log-domain Sinkhorn, and unbalanced optimal transport from first principles to evaluate competing attention-normalization schemes, collaborating with PhD researchers on experiment design

Undergraduate Researcher – Cybersecurity & Embedded Systems

Sep 2025 – Present

American University of Sharjah

Sharjah, UAE

- Benchmark Ascon-based lightweight cryptography on Raspberry Pi under Dr. Anastassia Gharib, comparing 3+ variants across execution time, memory footprint, and security parameters to surface the most efficient configuration for resource-constrained IoT hardware
- Lead author of an in-progress research paper on lightweight cryptography for IoT security in a 2-person research team

Co-Founder, President & Vice President

Jan 2026 – Aug 2026

AUS Blockchain Club

Sharjah, UAE

- Co-founded AUS's first blockchain club and grew membership to 15+ in its first semester; launched 3 technical sessions covering smart contract development and Web3 fundamentals
- As President, owned the club budget, ran operations and recurring member meetings, and set strategic direction for the officer team
- Organized a hands-on workshop on building a cryptocurrency from scratch attended by 20+ students

Course Grader – Web Applications & Development

Jan 2026 – May 2026

American University of Sharjah

Sharjah, UAE

- Graded 60+ student assignments across Java, JavaScript, SQL, HTML, and CSS, providing written technical feedback on implementation issues, logic errors, and code quality in full-stack coursework

Quality Assurance Officer

Sep 2025 – Dec 2025

AUS Cyber Security Club

Sharjah, UAE

- Reviewed and edited 10+ workshop materials and cybersecurity presentations across 3 club events, and delivered 2 workshops to 20+ attendees end-to-end

Projects

AUS QuickServe | GitHub

1st Place / Best Project

- Built a team-based food ordering web application using HTML, CSS, JavaScript, Java, and MySQL, with user login, order placement, receipt generation, and role-based flows for students and faculty
- Led database and backend development end-to-end; project placed 1st out of 12 teams on presentation day

Offline Document Summarization Tool

- Built a Python CLI tool that summarizes multi-page PDFs fully offline via a local LLM (Jan AI), removing cloud dependency and keeping documents private while completing in under 120 seconds per document

Certifications

- IBM: Quantum Machine Learning | Jul 2026
- CITI Program: Mandatory RCR for Research Scholars | Mar 2026
- Simplilearn: CCNA 200-301 Exam Preparation
- Cisco: Linux Unhatched